Energy Budget and Time–to–Failure Analysis of Wi-Fi, LoRa, and 3G/4G Back-Hauls in a Landslide-Monitoring Node

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***Abstract*—Early-warning landslide networks must operate for extended periods without maintenance in inaccessible terrain where solar harvesters are often shaded by vegetation. This paper compares the energy performance of three back-haul technologies — 3G/4G (LTE Cat-M1), WiFi, and LoRa — embedded in identical soil-moisture/rainfall sensor nodes. Moreover, we investigate the Wake-Up on Radio (WuR) technology in conjunction with these three types of sensor devices. All nodes share a $520\,\mathrm{W\,h}$ usable $LiFePO_4$ battery ($12\,\mathrm{V}$, $62\,\mathrm{A\,h}$, 70 % depth-of-discharge). Average power was measured with an open-hardware *EneMeter* while emulating three duty-cycle regimes: (i) *Normal* (25 min sleep + 5 min upload), (ii) *Rain-on-set* (10 min sleep + 5 min upload), and (iii) *Vigilance* (5 min sleep + 5 min upload). We also consider a fourth regime (*Special-WuR*) where the sensor node is in *Vigilance* mode 10% of the time during the year and deep sleeping during the remaining period. Resulting time-to-failure (TTF) spans from ≈18.3 months for a LoRa node in normal mode down to ≈101 days for a 3G/4G node forced to run in *Vigilance*. Once WuR technology is applied, the sensor nodes wake up in *Vigilance* mode only when necessary, leading to an increase in TTF of more than $5\times$ on average, regardless of the type of sensor node. These insights advise hybrid-link orchestration, WSN topology, the strategic use of WuR technology, and battery sizing for resilient landslide-monitoring deployments.**

***Index Terms*—IoT, WSN, WuR, energy consumption model, landslide monitoring, disaster monitoring, wake-up on radio**

I. Introduction

Recent years have witnessed an unprecedented rise in natural disaster alerts in Brazil. In 2024, the National Center for Monitoring and Early Warning of Natural Disasters (Cemaden) issued 3620 alerts, the highest annual total since its inception in 2011. Of these, 47% relate to hydrological risks such as flash floods and river overflows, and 68% of the 1690 recorded disaster occurrences were hydrological, reflecting the recurring impacts of heavy rainfall on urban areas [1]. Globally, extreme rainfall triggers over 80% of catastrophic landslides [2].

Motivated by national and international imperatives, this work presents an analysis of a fundamental aspect of designing and implementing a distributed landslide monitoring system: the energy budget. Although most studies in the area of Wireless Sensor Networks (WSN) assume the employment of energy harvesting, *in situ* landslide monitoring sensors typically operate in areas of difficult access where the cost of regularly maintaining batteries and solar panels can be multiple times the price of the sensors themselves thus preventing the pervasive use of such systems when compared to other environmental monitoring solutions.

The primary objective of this study is to cross-compare the energy consumption of multiple data communication solutions for landslide monitoring applications operating under different duty cycle regimes, given that most previous studies restrict themselves to evaluating a single protocol in isolation. We estimate the time-to-failure (TTF) of three variants of battery-powered nodes that start with the same initial energy capacity, without charging their batteries until each node reaches energy starvation and stops operating. To this end, we have developed a reliable energy model that supports most of the hardware sensor platforms currently adopted for environmental monitoring. In addition, we also evaluate Wake-up on Radio technology (WuR) [3] [4], showing significant improvements in TTF when using it coexisting with any protocol as well as any duty cycle regime under consideration.

II. Methodology & Assumptions

Both the technical specifications of commercial wireless sensor nodes and the WSN research literature are rich sources of information regarding the energy-budget topic. However, specific implementation details may still be missing, leading to inaccurate energy estimations, such as the transient effects during the power-cycling of the electronic components, voltage level variations at the power source (e.g., the battery), and the type and efficiency of voltage regulators [5]. In this Section, we develop a semi-empirical model that can capture these details and does not depend on the availability of an energy model for the WSN nodes under investigation.

A. Semi-empirical Energy Model: Part I

The proposed energy model starts by defining what the expected operational modes of the node are and how frequently they occur during a specific period (e.g., one day).

Next, for each of these modes, a low-cost instrument called EneMeter [6] is used to measure the power consumption (note: we developed EneMeter to be code/schematics open-source, employing the commercial chip LTC2943 from Linear Technologies). Even with a small sampling rate of 25 samples per second (SPS), we can improve the low-cost EneMeter accuracy significantly by taking multiple measurements over a long period, such as 24 hours, and storing them in an SD card for further processing. Fig.1 shows how EneMeter is connected to the sensor node to take power consumption measurements. During a period t^x $[s]$ while the node is under operational mode x, the integral over time of the curve representing measured voltage-$[V]$ multiplied by current-$[A]$ (i.e., $pwr^x_{inst.}$) represents the ene^x_{avg} $[W.s = J]$, which is the average energy consumed when the node operates in mode x during a period t^x. The accuracy of these calculations depends on how many measurements $cycles$ with periods t^x are taken to estimate ene^x_{avg}. Even if the period t^x varies, the model can still be accurate if we take enough measurements to properly represent this variability by averaging out all calculations for ene^x_{avg}.

B. Operational Modes of a Landslide Monitoring Sensor Node

We evaluate the energy consumption of a few communication protocols suitable for environmental monitoring applications. We have evaluated three commercial hardware options for landslide monitoring, and for each platform, we assess three distinct operational modes to analyze the tradeoffs between data quality and energy management. For extreme cases in terms of energy budget, we assume that the nodes operate without any dynamic change of their operational modes and *without any battery recharging*. Such constraints allow the investigator to estimate how long a sensor node can operate without local maintenance.

We regard nodes' configurations with distinct duty-cycles that emulate four operational modes concerning sensing, packetization, and link activation:

- **Normal cycle (A):** 5 min sense/upload $+25$ min sleep.
- **Rain-on-set (B):** 5 min sense/upload $+10$ min sleep.
- **Vigilance (C):** 5 min sense/upload $+5$ min sleep.
- **Special-WuR (D):** 10% of the time in *Vigilance* mode; remaining time in deep-sleep (i.e., hibernation).

Transitions between the above modes can potentially occur depending on the accumulation of rainfall. For instance, $10\ \mathrm{mm}/3\ \mathrm{h}$ may trigger the transition A→B, and $30\ \mathrm{mm}/3\ \mathrm{h}$ the transition B→C. However, for this study, which focuses on extreme cases, we do not handle inter-mode transitions for mode D. All single transmissions concern a fixed 100-byte MQTT payload per uplink, ensuring the energy figures are comparable across protocols. As will be discussed later, the results indicate that WuR is a strategic technology worthy of further investigation for disaster-monitoring deployments.

C. Semi-empirical Energy Model: Part II

A useful energy model for WSNs supports scenarios where the nodes are regularly changing their energy profiles. For instance, we envision that a WSN for landslide monitoring will be intensively sampling after a long period of precipitation or, otherwise, taking very few measurements during a dry day. However, the frequency and duration of such events change according to the region, season, and other environmental conditions. Therefore, for such applications, even if the hardware and software components are identical at two distinct sites, their energy models are also expected to be different. The proposition in this work is to observe the phenomenon (i.e., by taking power measurements) for a long period, such as one or more months, and analyze these measurements in terms of events per day with the proper energy characterization of each event. For instance, assume that for the previous year, it is known that a certain region had a total of $189h$ of heavy precipitation (e.g., matching with the *Vigilance* operating mode C). This period corresponds to one daily event C ($n_C = 1$) with a duration of around 78.75 $[s]$ per event (t^C_{event} = 78.75 $[s]$). If the average power consumed (measured) per event is $pwr^x_{avg} = 1798.45$ $[mW]$, one can estimate $ene^C_{avg-day} = 141.6$ $[J]$. Similarly, we can repeat this process for all remaining operational modes x of the sensor node by calculating their $ene^x_{avg-day}$ to reach the total energy consumed by the node each day. Moreover, we can take measurements for shorter periods (e.g., ten minutes) and extrapolate the calculation of $ene^x_{avg-day}$, although with a smaller accuracy.

Even if the characteristics of an operational mode of a node, such as duration and consumed power, are not constant during the period of interest, the proposed model can still be accurate as long as the number of observations is statistically sufficient to properly represent the variability of that mode during a long period of observation. The previous example considered an event (that is, heavy rain) that may not occur every day; nonetheless, it is appropriate to perform the analysis using at least one event per day since the energy model aims to have an output in terms of the remaining number of *days* that we expect the node to be operational. This *daily* approach significantly helps if one also monitors any form of energy harvesting that one wishes to include in that enhanced energy model. However, if an operating mode x of the sensor occurs multiple times a day, it is more accurate to specify this. One parameter of the model is n_X, the *average* number of times per day that one expects the node operating at mode X. While mathematically there is no difference on using $n_X = 1$ and a longer t^x for the entire day compared to using $n_X = 144$ and a smaller t^X, the former approach is a better choice if one wants to enhance this proposed semi-empirical model by adding energy *penalties* (e.g., due to electronic transient effects) whenever the node changes its operating mode. The following energy model was developed, where m is the *number of operating modes*, t^X in [s] and pwr^X_{avg} in [W]:

$$ene_{avg-day}\ [J]/[day]\ = \sum_{i=1}^{m} ene^i_{avg-day} \tag{1}$$

$$ene_{avg-day}^{X}\ [J]/[day]\ = n_X \times t^X \times pwr_{avg}^{X} \tag{2}$$

For the extreme cases to be discussed in this study (i.e., continuous operation under modes A/B/C/D until battery depletion), we simplify the model by adopting Eq.(2) and considering $t^X = 24 \times 3600[s]$ and $n_X = 1$:

$$ene_{avg-day}^{X}\ [J]/[day]\ = 86400 \times pwr_{avg}^{X} \tag{3}$$

D. Hardware Setup

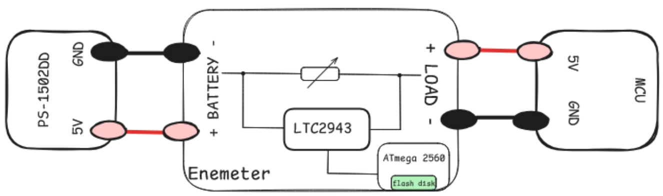

Fig. 1. Power measurements. A single 5 V rail from the Power-supply (PS-1502DD) passes through EneMeter's 315 mΩ Kelvin shunt before reaching the sensor node. The LTC2943 samples $V(t)$ and $I(t)$ at 1 kSPS but averaged data frames are streamed to SD card at 20 SPS.

Radio nodes: 3 firmware-identical sensor nodes:

- LoRa: ESP32-S3 MCU with SX1262 transceiver.
- Wi-Fi: ESP32-WROOM-32E module.
- 3G/4G: ESP32-WROOM-32 with SIM7000G modem.

Power source: During sampling, we do not employ any battery. Instead, nodes receive energy from a bench power supply (PS-1502DD) set to 5.000 V with a 2 A current limit. For real outdoors applications, typically a 12 V battery is involved, which changes its voltage level as it discharges. However, to allow a fair comparison between different experiments, we eliminated the impact of the battery by using a power supply, and we accounted for some effects by using a reduction factor of 30% when estimating the TTF for a sensor node.

EneMeter : This is an open-hardware, 315 mΩ shunt instrumented by an LTC2943, power-monitor device (code and CAD at [6]). The microcontroller unit (MCU), ATMega2560, commands the LTC2943 to take continuous measurements and send them to the MCU via I^2C interface. The MCU averages data to transfer to the SD card at the 20 SPS rate.

Measurement procedure: Each node cycles through the four duty profiles (A, B, C, and D), and each profile includes active and sleep modes, as shown in Fig.2. Six 5-minute slices (radio × state) form the dataset analysed in Section III. For every 5-minute slice we captured (i) a full-bandwidth $V(t), I(t)$ trace at 50 ms resolution, (ii) a JSON summary (mean, RMS, peaks, dwell times), and (iii) the raw MQTT traffic for replay. Since this study does not consider transitions between operational modes A, B, and C, we use Eq.(2) to calculate the daily energy consumption values shown in Table I. After this phase, we explore duty profile D - the only additional step was to measure the average power consumption of the WuR receiver connected to one of the sensor nodes (for more details of this WuR device see [3]).

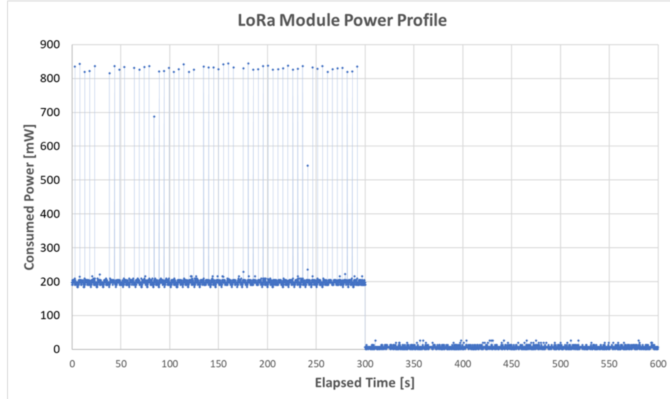

Fig. 2. Power profile of the LoRa node: sense/upload (left) and sleep (right).

E. Time-to-Failure Estimation

To estimate the time-to-failure (TTF) assuming no transition among operational modes A, B, and C, or considering mode D, we employ Eq.(4) with the value of $ene_{avg-day}^{X}$ provided by Eq.(3). The term X refers to the identification of the case under consideration, that is, the node type and its operational mode (A, B, C, or D):

$$TTF^X\ [days]\ = \frac{E_{batt}}{ene_{avg-day}^{X}} \qquad \frac{[J]}{[J]/day} \tag{4}$$

E_{batt} is the estimated initial energy capacity of the battery, which is assumed not to be rechargeable. Typically, the capacity specification of a battery is in Ah. In this case, it is necessary to use the nominal voltage level of the battery to estimate E_{batt} in $[J]$. The following is an example we consider in this work ($12V, 62Ah$ battery), using a reduction-factor of 30% mainly to account for the fact that not all initial energy of the battery can be used (also note that $1V \times 1A = 1W$, and $1W \times 1s = 1J$):

$$E_{batt} = (70\%) \times 12\ [V] \times 62[A][h] \times 3600\frac{[s]}{[h]} = 1874.9[kJ] \tag{5}$$

While $ene_{avg-day}^{X}$ can be estimated using additional theoretical energy models for the sensor node analysis, this study highlights the importance of empirically confirming the accuracy of such models by measuring the energy with an instrument such as the EneMeter. Mathematically, we achieve this by integrating measured voltage $[V]$ with measured current $[A]$ over time. EneMeter provides the average of instantaneous power for each sampling period of almost $50ms$. Next, the instrument sums up each quantized energy level. This method is more accurate than simply measuring current and voltage levels and averaging them to achieve an apparent average power, in particular when rapid transients are involved. After calculating the accumulated consumed energy for a sensor mode X, one can divide that value by the period necessary for measurements, thus estimating an accurate average power for that mode X, as shown in Table I.

III. Results and Discussion

As shown in Table I, we find that the smallest average power consumption per cycle is $39.5mW$, which corresponds to the LoRa node operating at mode A (i.e., sensing/uploading for 5 minutes and sleeping for 25 minutes). While individual components of a sensor node can still reach sleep power on the order of a few μW, it isn't easy to maintain the same energy efficiency for the whole sensor node. Internal voltage regulators of the sensor nodes are typically responsible for this mismatch [5]. While the LoRa node is more energy-efficient compared to WiFi and 4G, that advantage fades away when we introduce the WuR technology. The operational mode D (i.e., the particular WuR mode with the node hibernating 90% of the time) excels (Table II): even the power-hungry 4G sensor node can have a lifetime of almost 2 years without battery recharging and operating in *vigilance* mode whenever is necessary (constrained to $\approx$ 1617 accumulated hours in 2 years). Note that we assume the existence of a special node that can wake up the other nodes when precipitation occurs.

TABLE I
Average *cycle* power and daily energy (EneMeter measures)

Mode	Wi-Fi		LoRa		3G/4G	
	P_{avg} [W]	E_{day} [kJ]	P_{avg} [W]	E_{day} [kJ]	P_{avg} [W]	E_{day} [kJ]
Normal	0.0590	5.09	0.0395	3.41	0.0715	6.18
Rain-on-set	0.1029	8.89	0.0725	6.26	0.1427	12.33
Vigilance	0.1469	12.69	0.1055	9.12	0.2139	18.48
90% WuR	0.0256	2.21	0.0215	1.86	0.0322	2.78

One of the main challenges in landslide *in situ* monitoring applications is the cost of maintaining such systems. While solar panels are widely used and available, such components indeed require maintenance for periods that can be shorter than one year. In this study, we estimated the TTF (Table II) of such sensing systems assuming a typical battery (with the same physical volume as the auto industry batteries), without recharging. The results of the proposed semi-empirical model show that while the selection of a low-power radio technology, such as LoRa, can significantly improve TTF, the most important finding in this study is that the Wake-up on Radio (WuR) technology is even more critical for disaster monitoring applications in general.

TABLE II
Time-to-failure [days] with a 520 Wh (1874.9 kJ) battery.

Mode	Wi-Fi	LoRa	3G/4G
Normal	368	550	303
Rain-on-set	211	299	152
Vigilance	148	206	101
Special-WuR	849	1010	674

While WuR technology slightly increases the cost of the sensor node, it might well be a small price to pay for keeping a more reliable system. Moreover, the introduction of WuR with the capability of long-range communication has the potential to increase the reliability of such critical monitoring applications.

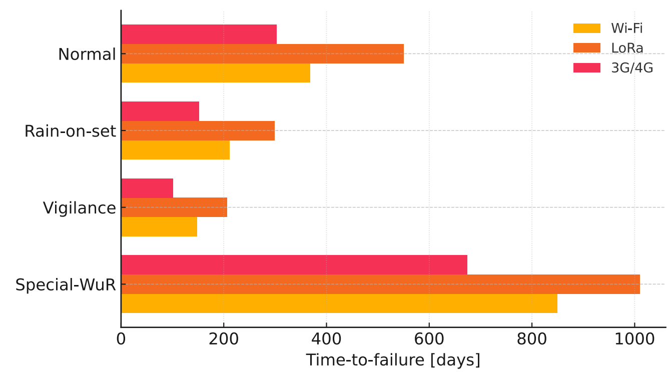

Fig. 3. Estimated autonomy for each back-haul under the three duty-cycles.

IV. Conclusion

The ultimate goal of this work is to contribute to efforts to model the energy budget of a WSN designed explicitly for disaster monitoring applications, such as a solution that provides warnings of an imminent landslide. This investigation concluded that sensor nodes without a solar panel can withstand operation for more than 2 years without battery replacement, and we have noticed that the WuR technology is a key changing factor. In practice, a rainfall-aware two-threshold scheduler — LoRa for routine telemetry, WiFi for moderate events, and LTE-M for critical alarms — further amplifies autonomy while preserving rapid responsiveness. As a future work, we intend to add the *positive energy* (battery recharging) to change the necessary maintenance cycles to multiple years.

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